

Basic SQL SELECT

Project Report

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Organisation	Veda Technology
Track	Data Analytics
Task	Basic SQL SELECT (SELECT, WHERE, ORDER BY)
Dataset	Chinook (digital music store)
Tools	PostgreSQL (pgAdmin 4); MySQL-compatible script also provided
Date	28 September 2026

1. Objective

The goal of this task is to build SQL fundamentals by writing simple queries that use SELECT, WHERE and ORDER BY. Ten queries were written on the Chinook sample database, each highlighting a different filtering or sorting technique, and the outputs were recorded.

2. Dataset Overview

Chinook models a small digital music store. Four of its tables were used in this task:

Table	Rows	What it stores
customer	59	Customer name, company, city, country, email
invoice	412	Purchases: date, billing country, total
track	3,503	Songs: name, composer, length (ms), price
employee	8	Staff: title, hire date, reporting line

3. Tools and Setup

- PostgreSQL with pgAdmin 4 was used to run the queries; a MySQL version of the load script is also provided.
- The Chinook data was converted into chinook_postgres.sql (CREATE TABLE + INSERT statements) and loaded into a database named chinook.
- All table and column names are lowercase and unquoted, so the same queries run on both PostgreSQL and MySQL.

4. Queries and Outputs

Each query is shown with its purpose, the SQL, and a sample of the output (first rows only; the full outputs are in query_outputs.txt).

Q1. Select specific columns

View name, country and email of all customers (SELECT with chosen columns).

```
SELECT customerid, firstname, lastname, country, email
FROM customer;
```

customerid	firstname	lastname	country	email
1	Luís	Gonçalves	Brazil	luisg@embraer.com.br
2	Leonie	Köhler	Germany	leonekohler@surfeu.de
3	François	Tremblay	Canada	ftremblay@gmail.com
4	Bjørn	Hansen	Norway	bjorn.hansen@yahoo.no
5	František	Wichterlová	Czech Republic	frantisekw@jetbrains.com
6	Helena	Holý	Czech Republic	hholy@gmail.com

59 row(s) returned - first 6 shown

Q2. Unique values with DISTINCT + ORDER BY

List every country where customers live, sorted A to Z.

```
SELECT DISTINCT country
FROM customer
ORDER BY country;
```

country
Argentina
Australia
Austria
Belgium
Brazil
Canada

24 row(s) returned - first 6 shown

Q3. Simple WHERE filter

Find all customers who live in Brazil.

```
SELECT customerid, firstname, lastname, city, email
FROM customer
WHERE country = 'Brazil';
```

customerid	firstname	lastname	city	email
1	Luís	Gonçalves	São José dos Campos	luisg@embraer.com.br
10	Eduardo	Martins	São Paulo	eduardo@woodstock.com.br
11	Alexandre	Rocha	São Paulo	alero@uol.com.br
12	Roberto	Almeida	Rio de Janeiro	roberto.almeida@riotur.gov.br
13	Fernanda	Ramos	Brasília	fernadaramos4@uol.com.br

5 row(s) returned

Q4. WHERE with IN + multi-column ORDER BY

Customers from Canada, France or Germany, sorted by country and then last name.

```
SELECT firstname, lastname, city, country
FROM customer
WHERE country IN ('Canada', 'France', 'Germany')
ORDER BY country, lastname;
```

firstname	lastname	city	country
Robert	Brown	Toronto	Canada
Edward	Francis	Ottawa	Canada
Aaron	Mitchell	Winnipeg	Canada
Jennifer	Peterson	Vancouver	Canada
Mark	Philips	Edmonton	Canada
Martha	Silk	Halifax	Canada

17 row(s) returned - first 6 shown

Q5. IS NOT NULL

Customers who have a company name recorded, sorted by company.

```
SELECT firstname, lastname, company, country
FROM customer
WHERE company IS NOT NULL
ORDER BY company;
```

firstname	lastname	company	country
Tim	Goyer	Apple Inc.	USA
Alexandre	Rocha	Banco do Brasil S.A.	Brazil
Luís	Gonçalves	Embraer - Empresa Brasileira de Aeronáutica S.A.	Brazil
Frank	Harris	Google Inc.	USA
František	Wichterlová	JetBrains s.r.o.	Czech Republic
Jack	Smith	Microsoft Corporation	USA

10 row(s) returned - first 6 shown

Q6. Pattern search with LIKE

Tracks whose name contains the word 'love' (case-insensitive using LOWER).

```
SELECT trackid, name, composer, unitprice
FROM track
WHERE LOWER(name) LIKE '%love%'
ORDER BY trackid;
```

trackid	name	composer	unitprice
24	Love In An Elevator	Steven Tyler, Joe Perry	0.99
56	Love, Hate, Love	Jerry Cantrell, Layne Staley	0.99
195	Let Me Love You Baby	Willie Dixon	0.99
335	My Love	Jauperi/Zeu Góes	0.99
341	The Girl I Love She Got Long Black Wavy Hair	Jimmy Page/John Bonham/John Estes/John Paul Jones/Robert Plant	0.99
345	Whole Lotta Love	Jimmy Page/John Bonham/John Paul Jones/Robert Plant/Willie Dixon	0.99

114 row(s) returned - first 6 shown

Q7. WHERE + ORDER BY DESC + LIMIT

Top 10 longest tracks (longer than 5 minutes), duration shown in minutes.

```
SELECT name, ROUND(milliseconds / 60000.0, 2) AS minutes
FROM track
WHERE milliseconds > 300000
ORDER BY milliseconds DESC
LIMIT 10;
```

name	minutes
Occupation / Precipice	88.12
Through a Looking Glass	84.81
Greetings from Earth, Pt. 1	49.34
The Man With Nine Lives	49.28
Battlestar Galactica, Pt. 2	49.27
Battlestar Galactica, Pt. 1	49.21

10 row(s) returned - first 6 shown

Q8. BETWEEN + ORDER BY

Invoices with total between 10 and 15, highest total first.

```
SELECT invoiceid, customerid, invoicedate, billingcountry, total
FROM invoice
WHERE total BETWEEN 10 AND 15
ORDER BY total DESC, invoiceid;
```

invoiceid	customerid	invoicedate	billingcountry	total
193	37	2011-04-23 00:00:00	Germany	14.91
5	23	2009-01-11 00:00:00	USA	13.86
12	2	2009-02-11 00:00:00	Germany	13.86
19	40	2009-03-14 00:00:00	France	13.86
26	19	2009-04-14 00:00:00	USA	13.86
33	57	2009-05-15 00:00:00	Chile	13.86

53 row(s) returned - first 6 shown

Q9. Multiple conditions with AND (date range)

USA invoices billed during the year 2010, latest first.

```
SELECT invoiceid, customerid, invoicedate, billingcity, total
FROM invoice
WHERE billingcountry = 'USA'
  AND invoicedate >= '2010-01-01'
  AND invoicedate < '2011-01-01'
ORDER BY invoicedate DESC;
```

invoiceid	customerid	invoicedate	billingcity	total
158	24	2010-11-19 00:00:00	Chicago	8.91
157	18	2010-11-16 00:00:00	New York	5.94
145	16	2010-09-23 00:00:00	Mountain View	13.86
137	28	2010-08-18 00:00:00	Salt Lake City	8.91
136	22	2010-08-15 00:00:00	Orlando	5.94
135	18	2010-08-14 00:00:00	New York	3.96

18 row(s) returned - first 6 shown

Q10. WHERE + LIKE + ORDER BY on dates

Sales-related employees ordered by hire date (oldest first).

```
SELECT employeeid, firstname, lastname, title, hiredate
FROM employee
WHERE title LIKE 'Sales%'
ORDER BY hiredate;
```

employeeid	firstname	lastname	title	hiredate
3	Jane	Peacock	Sales Support Agent	2002-04-01 00:00:00
2	Nancy	Edwards	Sales Manager	2002-05-01 00:00:00
4	Margaret	Park	Sales Support Agent	2003-05-03 00:00:00
5	Steve	Johnson	Sales Support Agent	2003-10-17 00:00:00

4 row(s) returned

5. Observations

- Customers are spread across 24 countries; 5 of the 59 customers live in Brazil.
- 17 customers come from Canada, France and Germany combined; 10 customers have a company recorded.
- 114 tracks contain the word "love" in their name.
- The longest track is "Occupation / Precipice" at 88.12 minutes.
- 53 invoices fall between 10 and 15 in total value; 18 USA invoices were billed in 2010.
- 4 employees have a sales-related title.

6. Learnings and Challenges

- SELECT with chosen columns is faster and cleaner than SELECT *.
- LIKE is case-sensitive in PostgreSQL, so LOWER(name) LIKE '%love%' was used to get consistent results on both databases.
- Row order can differ slightly between databases when sorting text (collation), so ties were broken with an extra ORDER BY column.
- Date ranges are safest written as >= start and < next start rather than BETWEEN.

7. Conclusion

This task covered the core building blocks of SQL querying: choosing columns, removing duplicates, filtering with comparison, IN, BETWEEN, LIKE and NULL checks, combining conditions with AND, and sorting or limiting results. These form the base for later topics such as aggregation and joins.